



Air University, Islamabad

Department of Creative Technologies

Project Proposal

for

SheAlert

*An Automatic Women Safety System Using
Sensors, Audio, and Camera*

Version 2.1 – Dataset Expanded

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Abbreviations and Document Conventions

The following identifier prefixes are used throughout this proposal to make objectives, requirements, and limitations easy to reference during evaluation and traceability checks.

Prefix	Stands For	Used In
OB-<i>n</i>	Project Objective	Section 4.1 (Objectives). Each objective is independently measurable.
FR-<i>m.n</i>	Functional Requirement (per module)	Section 8 (Modules). Numbered as <i>Module.Feature</i> (e.g., FR-3.2).
LM-<i>n</i>	System Limitation / Constraint	Section 9 (Limitations and Constraints).
Ref. [<i>n</i>]	Numeric IEEE-style citation	Inline references to academic literature; full list in Section 16.

Project Category

The following domains have been selected for this project (selected categories shown with a filled blue badge; unselected shown in grey):

- ✓ **C** Problem Solving & Artificial Intelligence
✓ **E** Smartphone Application
- ✓ **H** Image Processing
A Desktop Application
- B** Web Application
D Simulation and Modeling
- F** Smartphone Game
G Networks

1. Abstract

Women in Pakistan have no safety tool that works without their active involvement, and that gap becomes life-threatening the moment someone is grabbed or restrained and cannot reach their phone. Apps like bSafe, Himmat Plus, and ResQ Pakistan all share the same design flaw: they wait for a button press that may never come [6]. The few gesture-based options on the market still assume the user retains physical control, which the trauma literature shows is often not the case in real assault situations [15]. The core challenge, therefore, is building a

system that detects danger without any deliberate input from the user, works in Urdu acoustic environments where English-trained classifiers fail, and stays light enough on battery to run all day. No published system combines passive multi-modal sensing with a localised South Asian audio model on a standard Android device.

SheAlert addresses this by fusing a supervised multiclass LSTM motion classifier — replacing the originally proposed anomaly-detection autoencoder — with a YAMNet model fine-tuned on a purpose-built Urdu distress corpus, connected through a context-adaptive fusion engine and a two-stage cancellation window that cuts false alarms before any SOS is sent. The motion model classifies four activity classes (Normal, Fall/Collapse, Struggle/Resistance, Forced Movement), directly eliminating the false-positive problem inherent in anomaly detection. The audio pipeline incorporates a dedicated noise cancellation pre-processing stage before classification to improve robustness in real-world acoustic environments. The audio corpus combines 800–1,000 original Urdu distress recordings with three peer-reviewed Urdu and cross-lingual corpora: UrduSER [20] (3,500 clips, CC BY 4.0), UrduSpeech [21] (filtered subset of 59.2 h, CC BY 4.0), and SEMOUR [22] (15,040 clips, studio-quality Pakistani actors), together with public supplementary datasets, yielding a total pre-augmentation corpus of approximately 14,000–16,000 clips. This expanded multi-corpus strategy directly addresses Panel Comment 2 on dataset diversity by combining broadcast Urdu (UrduSER), large-scale naturalistic speech (UrduSpeech), and controlled studio recordings (SEMOUR), ensuring YAMNet is exposed to the full acoustic range of Pakistani Urdu emotion before domain adaptation. Model performance will be evaluated via per-class F1-score using 5-fold speaker-stratified cross-validation; score targets will be defined after implementation based on dataset characteristics, as each dataset has its own standard and no specific threshold is pre-committed. Battery overconsumption is treated as an anticipated risk: continuous background sensor and audio processing is expected to increase battery drain, and the three-tier cascaded pipeline with noise cancellation is the planned mitigation, benchmarked in Month 1 across three configurations and three device tiers. Beyond the app itself, the Urdu assault-distress dataset will be released publicly and represents the first labelled resource targeting assault-specific Urdu distress vocabulary, making it a standalone research contribution.

2. Introduction

This proposal describes *SheAlert*, a Final Year Project in the Department of Creative Technologies at Air University, Islamabad, built to address a straightforward but largely ignored problem: women's safety apps do not work when they are needed most. Reviews of the field consistently find the same picture – a panic button, an SMS with a GPS link, sometimes a shake gesture – while the underlying sensing and machine learning capabilities of modern smartphones have moved far beyond what any of these apps actually use [6, 18]. Activity recognition on phone sensors is a solved problem [1, 12], audio event classification runs comfortably on-device [9, 17], and multi-modal fusion has a mature literature behind it [2, 3]. What does not exist, as far as this team has been able to determine, is a system that puts all of those pieces together into a fully passive safety pipeline that runs end-to-end on a standard Android handset. SheAlert is that system. It monitors motion and audio continuously in the background, fuses both streams into a live threat score, and sends a GPS-tagged SOS with an audio clip and optional camera

snapshot when the score crosses an empirically validated threshold and the user does not cancel within a 20-second confirmation window. The project is carried out by three students with clearly divided module ownership, under the supervision of Dr. Samana Batool, with Dr. Madiha Yousaf as Co-Supervisor and Maryam Aamir Ashraf as Industrial Supervisor.

3. Problem Statement

Women across Pakistan face frequent harassment, stalking, and assault in everyday public environments. Current technology offers no effective *passive* protection against these threats – every available solution requires a deliberate action from the user.

3.1 The Manual-Action Assumption

The fundamental flaw of existing applications is the assumption that a person in genuine danger retains the freedom, mental clarity, and device access required to operate a phone. The trauma literature on tonic immobility and peritraumatic freezing is fairly unambiguous on this point: a large fraction of assault victims report being unable to move or act in the moments around an attack [4, 15]. On top of that, attackers often take or knock away the phone before the victim can use it. Safety systems gated on a deliberate action therefore fail in precisely the situations they are sold for.

GPS sharing, which a few apps offer continuously, has a related problem. It tells a contact where the user is, not whether anything is wrong. Whoever receives the location has to decide for themselves whether it looks suspicious, and at 3am on a Tuesday that decision usually does not get made. No widely deployed system reads the phone's own sensors and forms a judgement about danger on its own.

3.2 The Language and Cultural Gap

The published audio-safety work we found is built on English-language datasets, mostly Google AudioSet [9]. Distress in Urdu, including the phrases people actually shout (*bachao*, *madad karo*, *choro mujhe*), has not been modelled or labelled in any safety-system dataset indexed by IEEE Xplore, the ACM Digital Library, arXiv, or Google Scholar at the time of this proposal. While UrduSER [20], UrduSpeech [21], and SEMOUR [22] provide broad Urdu emotional speech data, none contains assault-specific distress vocabulary, making SheAlert's original recording contribution the first of its kind for safety applications. For a population of roughly 241 million [16], an English-only audio classifier is going to miss the most common cues.

3.3 The Battery and Deployment Constraint

Battery overconsumption is an anticipated challenge for any always-on safety application. The mobile deep-learning literature [13, 17] is consistent that running audio inference continuously without optimisation can burn more than 15% of battery per hour on mid-range hardware, which would make the application unusable in practice. It is anticipated — though not yet confirmed — that SheAlert's continuous background sensor and audio monitoring will cause elevated battery drain compared to a standard idle application, since the background service must keep the

CPU active, poll motion sensors at 50 Hz, run the noise cancellation stage, and trigger audio inference whenever the VAD cascade fires. To mitigate this anticipated issue, SheAlert employs a 3-tier cascaded audio pipeline (Module 3) combined with a dedicated noise cancellation pre-processing stage (FR-3.0), designed to minimise unnecessary heavy inference. The actual battery consumption of this architecture is an empirical research question measured in Month 1 of the project; the production pipeline configuration will be selected based on those results and reported in the thesis.

3.4 Combined Gap Addressed by SheAlert

The three problems above are usually treated separately, but in practice they reinforce each other. A passive system that does not speak the user's language will miss real events. A language-aware system that drains the battery in three hours will be uninstalled. A battery-efficient, language-aware system that still needs a button press defeats the original point. SheAlert is designed to handle all three together on a standard Android phone with no extra equipment, and the rest of this document is about how the architecture makes that combination feasible.

4. Problem Solution and Objectives of the Proposed System

SheAlert proposes a fully passive Android application that monitors motion and audio continuously, fuses both streams into a real-time threat score, and transmits an emergency SOS automatically when the score exceeds a validated threshold and two sequential confirmation stages pass without cancellation.

The motion layer uses a supervised multiclass LSTM classifier [10] trained on UCI HAR [1], WISDM [12], and supplementary public fall datasets (MobiAct, UniMiB SHAR) together with original volunteer recordings, covering four threat-relevant classes: Normal Activity, Fall/Collapse, Struggle/Resistance, and Forced Movement. This supervised multiclass formulation directly addresses the false-positive problem of anomaly-detection approaches by training the model on labelled examples of every class it must distinguish. The audio layer includes a dedicated noise cancellation stage followed by YAMNet [8, 9] fine-tuned via a two-stage domain-adaptation strategy on an expanded multi-corpus foundation comprising UrduSER [20], UrduSpeech [21], SEMOUR [22], cross-lingual acoustic anchors (CREMA-D [5], RAVDESS), and the team's original Urdu distress recordings (Section 10). A 3-tier cascaded pipeline limits heavy inference activations during normal use.

The fusion engine combines sensor and audio confidence scores using context-adaptive weights [2, 3]. Its operating threshold is determined empirically through ROC-curve analysis [7] on a held-out evaluation set, not preset. A two-stage confirmation window (10 s silent + 10 s audible) with a dead man's switch PIN reduces false-alarm transmission. After confirmation, MobileNet SSD [17] performs an optional 2–3-second person check; fail-safe logic ensures SOS proceeds even if the camera is unavailable.

All inference runs on-device. SOS delivery uses Firebase Cloud Messaging as the primary channel with automatic SMS fallback. The complete system ships as a Flutter Android application

with three sensitivity modes, post-event feedback, and an anonymous incident heatmap.

4.1 Objectives

- OB-1** Build and evaluate a supervised multiclass LSTM classifier for passive motion threat detection using UCI HAR, WISDM, MobiAct, and UniMiB SHAR for the Normal and Fall classes, and original volunteer recordings for the Struggle/Resistance and Forced Movement classes. Classification performance will be evaluated using per-class F1-score on a subject-disjoint test set across all four classes (Normal Activity, Fall/Collapse, Struggle/Resistance, Forced Movement); score targets will be set after implementation and cross-validation based on dataset characteristics, as no specific threshold is pre-committed. The team will aim for the highest accuracy achievable on the assembled data. An LSTM Autoencoder is retained as a baseline comparison in the ablation study only.
- OB-2** Construct a Urdu distress audio corpus by combining three peer-reviewed Urdu emotional speech corpora as the primary foundation: UrduSER [20] (3,500 clips, 10 professional Pakistani actors, CC BY 4.0), UrduSpeech [21] (filtered subset from the US-Std 59.2 h split, CC BY 4.0, yielding approximately 2,000–3,000 clips after emotion-field filtering for fearful/anxious, angry, and neutral categories), and SEMOUR [22] (15,040 clips, 8 professional Pakistani actors, studio quality, CC non-commercial research, yielding approximately 5,640 relevant clips across Fearful, Anger, and Neutral classes). Together these three Urdu corpora span broadcast naturalistic speech (UrduSER), large-scale multi-domain naturalistic speech (UrduSpeech), and controlled studio recordings (SEMOUR), providing YAMNet with the full acoustic range of Pakistani Urdu emotion before domain adaptation. These are supplemented by cross-lingual acoustic anchors: CREMA-D [5] Fear and Anger subsets (~2,400 clips, 91 actors, CC BY-SA 4.0) and the RAVDESS audio-only angry/fearful subset (~200 clips, CC BY-NC 4.0), which teach YAMNet the cross-lingual acoustic signatures of fear and anger (pitch spikes, energy bursts, vocal tremor) during Stage 1. Ambient and environmental classes are provided by AudioSet scream subset (~600–800 clips), ESC-50, and UrbanSound8K (~800 clips). Original recordings of 800–1,000 Urdu clips targeting assault-specific distress vocabulary (*bachao*, *madad karo*, *choro mujhe*) absent from existing datasets provide the domain adaptation signal for Stage 2 fine-tuning. Augmentation is applied exclusively as a regularisation technique to prevent overfitting; speaker and environment diversity is achieved entirely through the original recordings and the three Urdu corpora, not through augmentation. Total pre-augmentation corpus: approximately 14,000–16,000 clips. Apply a dedicated noise cancellation stage to all audio prior to feature extraction. Fine-tune YAMNet via two-stage domain adaptation (Stage 1 on the full combined corpus; Stage 2 on original Urdu recordings only). Classification performance will be evaluated via per-class F1-score through 5-fold speaker-stratified cross-validation (mean F1 $\pm$ standard deviation reported) and out-of-distribution evaluation on one held-out acoustic environment not seen during training; score targets will be defined after cross-validation results are available, with

the generalisation gap reported explicitly in the thesis.

- OB-3** Design and implement a context-adaptive weighted fusion engine, with operating threshold validated empirically through ROC-curve analysis on a held-out evaluation set; full ROC figures included in the final thesis.
- OB-4** Implement a two-stage cancellation window with dead man's switch PIN mode, and measure its effect on false-alarm SOS transmissions through the ablation study described in Module 5 (FR-5.5). Target: at least 60% relative reduction in false-alarm SOS transmissions when comparing the full system (with confirmation window) against the same fusion pipeline without the window, on the identical held-out evaluation set.
- OB-5** Empirically benchmark battery consumption across three candidate audio pipeline configurations — (a) always-on full YAMNet inference, (b) two-tier cascade (VAD + YAMNet), (c) three-tier cascade (VAD + noise cancellation + rule-based filter + YAMNet) — on three Android device tiers (budget, mid-range, flagship) using Android Battery Historian. It is anticipated that continuous background monitoring will result in elevated battery consumption due to persistent CPU activity, 50 Hz sensor polling, noise cancellation processing, and periodic audio inference; the cascaded pipeline architecture with noise cancellation is the primary architectural mitigation for this anticipated issue. The production architecture will be selected based on empirical results, not on any pre-committed consumption figure. Target thresholds and acceptable limits will be defined after Month 1 measurement and reported in the thesis.
- OB-6** Integrate all modules into a working Flutter Android application and evaluate end-to-end detection latency, false-positive rate, and battery consumption against documented benchmarks.
- OB-7** Conduct an ablation study demonstrating that multi-modal sensor and audio fusion outperforms any single-modality baseline in both detection rate and false-positive rate.

5. Related System Analysis / Literature Review

This section reviews published work in four areas relevant to SheAlert: women's safety applications, smartphone-based human activity recognition, on-device audio event classification, and multi-modal fusion. It then surveys deployed competitor applications, and concludes with the dataset-search methodology that supports our claim of novelty for the Urdu distress dataset.

5.1 Women's Safety Applications

Chand and Paul [6] reviewed mobile women-safety applications and noted that nearly all rely on a manually triggered panic button, with little algorithmic intelligence. Yarrabothe and Thota [18] proposed Abhaya, an SOS application combining GPS and SMS but again gated on a manual press. Kumar and Lenin Fred [11] surveyed Indian-market safety apps and concluded that

none employ machine learning for automatic threat detection. The common limitation across all reviewed systems is the manual-action assumption discussed in Section 3.1, which the trauma literature on tonic immobility [4, 15] suggests is exactly the wrong assumption to build a safety system on.

5.2 Smartphone Human Activity Recognition

Smartphone HAR has matured into a well-studied problem. Anguita et al. [1] released UCI HAR with 30 subjects performing six daily activities, which has become a standard benchmark. Kwapisz et al. [12] introduced WISDM, focusing on activity recognition from a single accelerometer. Recurrent architectures, especially LSTM networks [10], have repeatedly outperformed classical models on these datasets. For fall detection, the MobiAct dataset [23] provides four fall types and eleven ADL types from 66 subjects recorded on a Samsung Galaxy S3, directly matching SheAlert’s Android sensor pipeline. UniMiB SHAR [24] provides 11,771 samples of eight fall types and nine ADL types from 30 subjects, also collected via Android smartphone. For the Struggle/Resistance class, a systematic search of IEEE Xplore, ACM DL, arXiv, and Kaggle using query terms (*struggle*, *grab*, *forced movement*, *physical assault*) combined with (*accelerometer*, *IMU*, *smartphone sensor*) returned no labelled public dataset; original volunteer recordings constitute the primary data contribution for this class and an original experimental contribution of Module 2.

5.3 Audio Event Classification on Mobile Devices

Hershey et al. [9] introduced large-scale audio event classification with the AudioSet dataset and CNN architectures, which underpin Google’s open-source YAMNet [8]. Sandler et al. [17] introduced MobileNetV2, an efficient CNN backbone for on-device inference. Lane et al. [13] reviewed the trade-offs of running deep learning continuously on mobile hardware and motivated cascaded “early-exit” designs, which we apply as the 3-tier audio pipeline in Module 3.

Three peer-reviewed Urdu speech datasets directly inform the audio training pipeline. Akhtar et al. [20] released UrduSER, the first comprehensive Urdu speech emotion dataset, comprising 3,500 clips from 10 professional Pakistani actors sourced from drama serials, with seven emotion classes validated by 100 native speakers at 94% inter-rater accuracy (CC BY 4.0). Attia et al. [21] released UrduSpeech, a large-scale multi-domain Urdu speech corpus of 156 hours covering news, podcasts, vlogs, and comedy, with paralinguistic emotion labels and Pakistani-accent-dominant speakers (CC BY 4.0, arXiv:2605.17846); the US-Std split (59.2 h) is used and filtered by the emotion field to extract fearful/anxious, angry, and neutral subsets relevant to SheAlert’s classification taxonomy. Zaheer et al. [22] released SEMOUR, a phonetically balanced Urdu speech emotion dataset comprising 15,040 clips recorded by 8 professional Pakistani actors under studio conditions across 8 emotion classes; the Fearful, Anger, and Neutral classes together provide approximately 5,640 clips for SheAlert’s training corpus. Taken together, UrduSER, UrduSpeech, and SEMOUR span the acoustic range from broadcast naturalistic drama to large-scale multi-domain speech to controlled studio recordings, giving YAMNet comprehensive exposure to Pakistani Urdu emotional prosody before domain adaptation on assault-specific vocabulary. For cross-lingual acoustic grounding, CREMA-D [5] provides 7,442 clips from 91 actors across six emotions (CC BY-SA 4.0); the Fear and Anger subsets

(~2,400 clips) serve alongside RAVDESS to anchor YAMNet's Stage 1 representation of fear and anger acoustics prior to Urdu-specific fine-tuning.

5.4 Multi-Modal Fusion

Atrey et al. [2] provided a foundational taxonomy of multi-modal fusion strategies. Baltrušaitis et al. [3] surveyed multi-modal machine learning challenges including alignment, representation, and fusion, identifying weighted late fusion as a robust approach when modalities can fail independently. This is precisely the scenario in SheAlert, where the microphone may be obstructed but motion sensors continue to produce data. Threshold selection on the fused score is performed using standard ROC-curve methodology [7].

5.5 Deployed Competitor Applications

Beyond the academic literature, several deployed applications target women's safety. Table 1 summarises key weaknesses of representative systems and the corresponding solution offered by SheAlert.

Table 1: Comparison of Deployed Women's Safety Applications

Application	Weakness	SheAlert's Solution
bSafe / Shake2Safety	Requires the user to shake the phone or press a button. Fails when restrained or in shock. GPS sharing only, with no threat intelligence.	Passive AI detection with no user action required at any stage.
Himmat Plus (Delhi Police)	Manual SOS only. Tied to specific city infrastructure; cannot operate outside the registered municipal area.	On-device AI with no infrastructure dependency. Operates anywhere with mobile data or SMS.
Watch Over Me	Periodic check-in model. Fails if the user is incapacitated and cannot confirm safety.	Continuous passive monitoring with no check-in dependency.
ResQ Pakistan	Manual activation only. No AI component, no audio or motion analysis.	Multi-modal AI fusion with Urdu audio support tailored to the Pakistani context.

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Application	Weakness	SheAlert's Solution
Generic ML Safety Research [6, 18]	Existing ML-based proposals use English-only audio. No Urdu assault-distress dataset is indexed in IEEE Xplore, ACM DL, arXiv, or Google Scholar (search performed during this proposal).	SheAlert constructs and openly releases the first Urdu assault-distress dataset known to the authors, supplementing UrduSER [20], UrduSpeech [21], and SEMOUR [22] with assault-specific vocabulary.

5.6 Dataset Novelty: Search Methodology

To substantiate the claim that no Urdu assault-distress audio dataset is currently published, we conducted a systematic search of four databases (IEEE Xplore, ACM Digital Library, arXiv, and Google Scholar) using the query terms: (“Urdu”) AND (“distress” OR “scream” OR “panic” OR “women safety”) AND (“audio” OR “dataset” OR “classification”). The search returned UrduSER [20], UrduSpeech [21], and SEMOUR [22] as Urdu emotional speech corpora, but no dataset containing labelled assault-specific distress vocabulary (*bachao*, *madad karo*, *choro mujhe*) or targeted safety-system recordings. The construction and open release of such a dataset (Section 10, Module 4) is therefore a stated research contribution of this project, complementary to rather than duplicated by the above corpora, with the search methodology and date of search to be reported in the final thesis.

6. Vision Statement

For women in Pakistan and South Asia who face the routine possibility of harassment or assault in public spaces

Who need a phone that can call for help on their behalf when they are unable to act

The SheAlert Android application

Is a fully passive AI-powered personal safety system

That continuously monitors motion and audio in the background, detects threatening situations automatically, and sends an emergency alert containing GPS location, a short audio clip, and an optional camera snapshot to pre-registered contacts – without requiring any action from the user during a crisis

Unlike every other safety app available in Pakistan today, which all require the user to press a button, shake the phone, or hold a key before help is requested

Our product is built around the recognition that someone in real danger usually cannot do any of those things, so the phone in their pocket has to make the call instead.

7. Scope

SheAlert is an Android mobile application that runs continuously in the background and detects threatening situations automatically without user interaction during a crisis. The scope covers nine modules: motion sensing and multiclass threat classification, audio cascading and YAMNet classification, multi-modal fusion, two-stage confirmation, optional camera verification, SOS delivery, and the Flutter user-facing application.

The motion layer collects accelerometer and gyroscope data via the Android Sensor API, applies sliding-window segmentation, and runs a supervised multiclass LSTM classifier producing per-class threat scores. The audio layer implements a dedicated noise cancellation pre-processing stage (FR-3.0), followed by WebRTC voice activity detection (Tier 1), a rule-based acoustic filter (Tier 2), and YAMNet fine-tuned on an expanded multi-corpus Urdu foundation via two-stage domain adaptation (Tier 3). The audio corpus integrates three peer-reviewed Urdu corpora (UrduSER, UrduSpeech, SEMOUR), two cross-lingual acoustic anchor datasets (CREMA-D, RAVDESS), ambient datasets (AudioSet, ESC-50, UrbanSound8K), and the team's original Urdu distress recordings.

The fusion engine combines sensor and audio confidence scores into a weighted threat score with context-adaptive weights. Threshold selection is empirical via ROC-curve analysis on a held-out set. When the threshold is crossed, a two-stage cancellation window provides 10 seconds of silent and 10 seconds of audible cancellation opportunity. A dead man's switch PIN guards against attacker interference. After confirmation, MobileNet SSD performs an optional 2–3-second person check; fail-safe logic permits SOS regardless of camera availability.

The SOS payload includes GPS location from `FusedLocationProviderClient`, a 5-second audio clip, and an optional camera snapshot. Delivery is via Firebase Cloud Messaging with

SMS fallback. The Flutter application supports Safe, Standard, and Aggressive sensitivity modes; mandatory onboarding with contact registration; OEM-specific battery-optimiser whitelist prompts (Xiaomi, Samsung, Oppo, Huawei); persistent permission health monitoring; post-event feedback for per-user threshold personalisation; and an anonymous GPS-timestamped incident heatmap.

The scope explicitly excludes: iOS support, web interfaces, server-side AI processing, real-time video streaming, persistent local audio or video storage during normal monitoring, and clinical or legal certification as a personal safety device. A mandatory technical spike in the first month will validate continuous background microphone access across Android 10, 12, and 14 on three device tiers and benchmark battery consumption across three pipeline configurations before architecture finalisation.

Feasibility and workload distribution. Nine modules across three students is an ambitious scope, and it is worth being direct about how the team plans to handle it. Each student owns a coherent vertical slice: Mahad Jokhio covers the motion sensing and multiclass classification pipeline (Modules 1–2), Moaz Hassan Khan Manj covers the audio classification and fusion engine (Modules 3–5), and Hunain Ahmed covers system integration, the confirmation window, camera module, SOS delivery, and the Flutter application (Modules 6–9). This division means no single module is shared in a way that creates a dependency bottleneck, and each student has a deliverable they can begin independently from Month 1. The two highest-risk items are Module 4 (dataset collection) and Module 7 (camera verification). If volunteer recruitment for the Struggle class falls below 15 subjects by end of Month 2, the augmentation multiplier on collected clips is raised from $3\times$ to $5\times$ and the generalisability claim is explicitly narrowed to the recorded speaker demographics in the thesis. If Module 7 (camera) encounters persistent OEM compatibility issues, it will be demoted to an optional appendix contribution and the two-modality (sensor + audio) baseline will serve as the primary evaluated system. Supplementary public datasets (UrduSER, UrduSpeech, SEMOUR, CREMA-D, AudioSet scream subset, ESC-50, UrbanSound8K) are pre-downloaded in Month 1 as a buffer against recording delays. The core project deliverables — the multiclass LSTM motion model, the fine-tuned YAMNet classifier, the fusion engine, the confirmation window, and the SOS delivery system — remain completable even if both contingencies are triggered simultaneously.

8. Modules

8.1 Module 1: Motion Sensor Data Pipeline

Module 1 — Motion Sensor Data Pipeline

Owner: Mahad Jokhio (231242)

- FR-1.1** Continuous accelerometer and gyroscope collection via Android `SensorManager` and `SensorEventListener` at 50 Hz.
- FR-1.2** Real-time sliding-window segmentation: 2-second window with 50% overlap.
- FR-1.3** Per-window z-score normalisation using mean and standard deviation pre-computed during model training.
- FR-1.4** Normalised tensor streamed to Module 2 with target end-to-end pipeline latency below 20 ms per window.
- FR-1.5** Sensor hardware availability monitor; raises a flag to the fusion engine on hardware failure or API disconnection.

8.2 Module 2: Motion Threat Classification (Multiclass LSTM)

Module 2 — Motion Threat Classification (Multiclass LSTM)

Owner: Mahad Jokhio (231242)

- FR-2.1** Supervised multiclass LSTM classifier [10] trained on four classes: (a) **Normal Activity** — walking, running, sitting, standing, stairs, sourced from UCI HAR [1] and WISDM [12]; (b) **Fall/Collapse** — four fall types from MobiAct [23] (66 subjects, Android smartphone) and eight fall types from UniMiB SHAR [24] (30 subjects, Android smartphone), selected because both datasets match SheAlert's smartphone sensor pipeline; (c) **Struggle / Resistance** — original volunteer recordings of simulated assault scenarios (wrist grab, forced directional pull, push/stumble), 15–20 campus volunteers, no public dataset exists for this class (confirmed gap); (d) **Forced Movement** — subset from MobiAct's dynamic ADL classes combined with volunteer recordings of walking under physical constraint.
- FR-2.2** Per-window class-probability vector output (softmax) streamed to the fusion engine in Module 5 as a threat score on 0–1 scale.
- FR-2.3** Subject-disjoint train/test split mandatory: test-set subjects have zero overlap with training-set subjects. 5-fold cross-validation with subject-level stratification during development; mean F1 $\pm$ standard deviation across folds reported in the thesis. Score targets will be determined after initial cross-validation results are available; no specific threshold is pre-committed.
- FR-2.4** TensorFlow Lite deployment for real-time on-device inference; target latency below 30 ms per window on a mid-range Android device.
- FR-2.5** Sensor-only fallback mode delivering continuous class scores when the audio mod-

ule is interrupted or unavailable.

- FR-2.6** Placement variability study (original experiment): controlled evaluation of multiclass accuracy across pocket, bag, and hand placements, with results reported as a contribution in the individual thesis.
- FR-2.7** LSTM Autoencoder [14] retained as baseline comparison in ablation study only; per-class F1 of multiclass LSTM compared against autoencoder anomaly-detection rate to quantify the improvement in false-positive reduction.
- FR-2.8** Volunteer data collection for Struggle/Resistance class: simulated scenarios recorded on campus using Physics Toolbox Sensor Suite (Android, CSV export); written informed consent obtained; each session 10–15 minutes per volunteer. This class constitutes an original data contribution of Module 2, analogous to the Urdu dataset contribution in Module 4.

8.3 Module 3: Audio Pipeline and 3-Tier VAD Cascade

Module 3 — Audio Pipeline and 3-Tier VAD Cascade

Owner: Moaz Hassan Khan Manj (231168)

- FR-3.0** **Noise Cancellation Pre-Processing Stage** (applied before all downstream audio tiers): raw audio captured from the microphone passes through a noise cancellation step prior to any voice activity detection or classification inference. This stage applies spectral subtraction combined with a Wiener filter to suppress stationary background noise (traffic, HVAC, crowd chatter) prevalent in the six target acoustic environments. The cleaned signal is then forwarded to Tier 1. This stage serves two purposes: (1) improving YAMNet classification accuracy by presenting cleaner input features, and (2) reducing spurious Tier 2 acoustic filter activations caused by background noise, which in turn reduces unnecessary heavy inference and contributes to the anticipated battery consumption mitigation. Processing runs on the CPU using lightweight algorithms (spectral subtraction is $\mathcal{O}(N \log N)$ per frame) and adds negligible latency to the pipeline.
- FR-3.1** Android Foreground Service with `foregroundServiceType=microphone` as required on Android 14+.
- FR-3.2** Tier 1: WebRTC VAD with RMS energy threshold applied to the noise-cancelled signal; near-zero battery cost; chunks without voice are discarded immediately.
- FR-3.3** Tier 2: rule-based acoustic filter checking RMS amplitude $> 8,000$, fundamental pitch > 300 Hz, and zero-crossing rate > 0.3 on the noise-cancelled signal; at least two of three must match.
- FR-3.4** Tier 3: YAMNet ML inference [8, 9] triggered only when Tier 2 is satisfied. Cascaded design follows mobile early-exit principles [13]. Battery consumption of this cascade is anticipated to be significantly lower than always-on inference due to the noise

cancellation and cascaded gating; empirical benchmarking in Month 1 (see OB-5) will confirm or revise this expectation and no specific consumption figure is pre-committed.

FR-3.5 Microphone interruption detection raising an internal flag to the fusion engine if audio is lost for ≥ 5 consecutive seconds.

FR-3.6 Microphone active only outside home geofence: audio pipeline starts when GPS position is outside the user-set home radius and pauses automatically once she returns; location polled via `FusedLocationProviderClient` at low-power priority.

8.4 Module 4: Urdu Distress Dataset and YAMNet Classification

Module 4 — Urdu Dataset and YAMNet Classification

Owner: Moaz Hassan Khan Manj (231168)

FR-4.0 External public corpus integration (pre-fine-tuning foundation):

Primary Urdu emotional speech corpora (three sources):

- **UrduSER** [20]: 3,500 clips, 10 professional Pakistani actors, balanced gender, age range 20–65, sourced from drama serials, CC BY 4.0. Class mapping: Fear (~500 clips) → Distress; Angry (~500 clips) → Aggression; Neutral (~500 clips) → Normal. Provides broadcast naturalistic Urdu emotional speech diversity.
- **UrduSpeech** [21]: 156-hour multi-domain Urdu speech corpus (news, podcasts, vlogs, comedy), Pakistani-accent-dominant, CC BY 4.0. The US-Std split (59.2 h) is used; filtered by the emotion field to extract fearful/anxious → Distress, angry → Aggression, and podcast/vlog/comedy neutral clips → Normal, yielding approximately 2,000–3,000 relevant clips. Provides large-scale naturalistic multi-domain Urdu speech complementing the broadcast register of UrduSER.
- **SEMOUR** [22]: 15,040 clips from 8 professional Pakistani actors, studio quality, phonetically balanced, 8 emotion classes, CC non-commercial research. Class mapping: Fearful (~1,880 clips) → Distress; Anger (~1,880 clips) → Aggression; Neutral (~1,880 clips) → Normal (approximately 5,640 relevant clips total). Provides controlled studio recordings that complement the naturalistic registers of UrduSER and UrduSpeech.

Cross-lingual acoustic anchors (two sources):

- **CREMA-D** [5]: 7,442 clips, 91 actors, 6 emotions, CC BY-SA 4.0. Fear and Anger subsets (~2,400 clips) used to teach YAMNet the cross-lingual acoustic signatures of fear and anger (pitch spikes, energy bursts, vocal tremor) during Stage 1, prior to any Urdu-specific fine-tuning. Substantially expands RAVDESS coverage for this function.

- **RAVDESS** audio-only angry/fearful subset (~200 clips, CC BY-NC 4.0): supplements CREMA-D for cross-lingual acoustic grounding.

Ambient and environmental classes: AudioSet scream subset (~600–800 clips, language-agnostic screams) supplements the Distress class; ESC-50 and UrbanSound8K (~800 clips) supplement the Normal class with urban ambient backgrounds relevant to Pakistani deployment environments.

FR-4.1 Original recording of 800–1,000 Urdu audio clips across three classes: **Distress** (*bachao* ×5, *madad karo* ×5, *choro mujhe* ×5 = 15 distress clips per volunteer), **Aggression** (threatening Urdu speech, 5 clips per volunteer), and **Normal** (casual Urdu conversation, 5 clips per volunteer), from 20–30 volunteers across six real acoustic environments (quiet indoor, indoor ambient, outdoor quiet, outdoor busy, indoor reverberant, vehicle interior). All recordings are passed through the noise cancellation stage (FR-3.0) prior to feature extraction. Augmentation (noise addition, pitch shift, time stretch) applied to original clips only for regularisation, with augmentation explicitly stated not to substitute for speaker or environment diversity.

FR-4.2 Combined pre-augmentation corpus breakdown and total:

- UrduSER relevant clips: ~1,500
- UrduSpeech US-Std filtered: ~2,000–3,000
- SEMOUR relevant clips: ~5,640
- CREMA-D Fear + Anger: ~2,400
- RAVDESS audio-only: ~200
- AudioSet scream subset: ~700
- ESC-50 + UrbanSound8K: ~800
- Original Urdu recordings: ~800–1,000

Total pre-augmentation corpus: approximately 14,000–16,000 clips. After augmentation on original Urdu clips only (2–3×): approximately 16,000–19,000 total training items. Real diversity is provided by three complementary Urdu corpora (UrduSER, UrduSpeech, SEMOUR) spanning broadcast, naturalistic multi-domain, and studio registers, plus 20–30 original recording volunteers across six acoustic environments, not through augmentation.

FR-4.3 MFCC and Mel-spectrogram feature extraction using a Librosa-equivalent on-device pipeline, applied after noise cancellation.

FR-4.4 Two-stage YAMNet fine-tuning [8]: **Stage 1** — fine-tune on the full combined corpus (UrduSER + UrduSpeech + SEMOUR + CREMA-D + RAVDESS + AudioSet + ESC-50 + UrbanSound8K + original Urdu recordings) to build broad Urdu emotional speech representation, cross-lingual acoustic fear/anger signatures, and acoustic diversity across Pakistani environments; **Stage 2** — domain-adaptation fine-tuning on original Urdu recordings only to specialise for assault-specific distress vocabulary (*bachao*, *madad karo*, *choro mujhe*) and Pakistani urban acoustic conditions. Both stages evaluated separately and results reported as an incremental comparison

table in the thesis, providing evidence of domain adaptation benefit. Classification performance evaluated via per-class F1-score through 5-fold speaker-stratified cross-validation and out-of-distribution evaluation on one held-out acoustic environment; score targets will be defined from cross-validation results and reported in the thesis, not pre-committed.

- FR-4.5** TensorFlow Lite conversion and on-device deployment; original Urdu assault-distress dataset publicly released under CC-BY-4.0 on project completion, constituting the first labelled Urdu dataset targeting assault-scenario distress vocabulary.

8.5 Module 5: Multi-Modal Fusion Engine

Module 5 — Multi-Modal Fusion Engine

Owner: Moaz Hassan Khan Manj (231168)

- FR-5.1** Late-fusion formula [2, 3] combining sensor threat score s (from multiclass LSTM, highest non-Normal class probability) and audio confidence score a : $\text{threat} = W_1 s + W_2 a$, with $W_1 + W_2 = 1$.
- FR-5.2** Context-adaptive weights: W_1 increased in noisy environments (audio less reliable), W_2 increased in quiet environments. Weights tuned empirically.
- FR-5.3** Threshold selection via ROC-curve analysis [7] on a held-out set; full ROC figures included in the thesis.
- FR-5.4** Per-user threshold personalisation: after two weeks of post-event feedback, the threshold is adjusted per user based on false-alarm history.
- FR-5.5** **Ablation study protocol (formal ownership: Hunain Ahmed).** Four conditions evaluated on an identical held-out test set comprising both simulated threat scenarios and normal-life recordings: (a) sensor-only (multiclass LSTM), (b) audio-only (YAMNet), (c) sensor+audio fusion without confirmation window, (d) full system with confirmation window. Additional comparisons: (e) LSTM Autoencoder baseline vs. multiclass LSTM to quantify false-positive reduction; (f) pipeline without noise cancellation vs. pipeline with noise cancellation to quantify the contribution of FR-3.0 to classification accuracy and false-alarm rate. Comparison metrics: per-class F1, false-alarm rate, end-to-end latency. Objective OB-4's 60% target is measured as the relative reduction in false-alarm SOS transmissions between conditions (c) and (d).

8.6 Module 6: Two-Stage Confirmation Window

Module 6 — Two-Stage Confirmation Window (Human-in-the-Loop)*Owner: Hunain Ahmed (231156)*

- FR-6.1** Stage 1 (silent): 10-second countdown, minimum brightness, soft vibration, large CANCEL button.
- FR-6.2** Stage 2 (audible): 10-second countdown, loud beep, screen flash, CANCEL button, activated only if Stage 1 passes.
- FR-6.3** Dead man's switch PIN: a 4-digit PIN must be entered to confirm safety; an attacker cannot cancel without the PIN.
- FR-6.4** Researched and evaluated as a human-in-the-loop false-alarm reduction system; quantified contribution reported in the ablation study (FR-5.5).
- FR-6.5** Formal evaluation of usability under simulated stress conditions during user scenario testing (10–15 volunteers).

8.7 Module 7: Camera Module (Third-Modality Verification)**Module 7 — Camera Module***Owner: Hunain Ahmed (231156)*

- FR-7.1** Optional camera confirmation using MobileNet SSD [17] for 2–3 seconds after both confirmation stages pass.
- FR-7.2** Evaluated as a third-modality contribution to multi-modal fusion precision; ablation comparison of two-modality (sensor + audio) versus three-modality (sensor + audio + camera) precision is reported in the thesis.
- FR-7.3** Fail-safe logic: SOS transmission proceeds regardless of camera availability or hardware failure.
- FR-7.4** On-device TFLite inference; target inference time below 500 ms on a mid-range Android device.

8.8 Module 8: SOS and Alert Delivery System

Module 8 — SOS and Alert Delivery System

Owner: Hunain Ahmed (231156)

- FR-8.1** Alert packaging: GPS from `FusedLocationProviderClient`, 5-second audio clip from rolling buffer, optional camera snapshot.
- FR-8.2** Primary delivery via Firebase Cloud Messaging; each contact receives a one-tap acknowledgement.
- FR-8.3** Automatic SMS fallback via Android `SmsManager` if Firebase acknowledgement is not received within 60 seconds.
- FR-8.4** FastAPI Python backend for alert forwarding and contact relay; JWT-authenticated administrative endpoints.
- FR-8.5** Emergency contact registration guard: SOS cannot be initiated if no contacts are registered.

8.9 Module 9: Flutter App, UI, Heatmap and Feedback

Module 9 — Flutter App, UI, Heatmap and Feedback

Owner: Hunain Ahmed (231156)

- FR-9.1** Mandatory onboarding: emergency contact registration (≥ 1), OEM battery-optimiser whitelist prompts (MIUI, One UI, ColorOS, EMUI), Android 12+ microphone-indicator explanation.
- FR-9.2** Main dashboard: real-time threat-score gauge, monitoring status, sensitivity mode, contacts list.
- FR-9.3** Sensitivity modes: Safe (raised threshold), Standard (validated default), Aggressive (lowered threshold for high-risk contexts).
- FR-9.4** Persistent permission health monitor: critical permissions checked on every foreground event; persistent red notification if any are missing.
- FR-9.5** Post-event feedback: user classifies each alert as true or false; feeds the threshold personalisation in Module 5.
- FR-9.6** Anonymous GPS-timestamped incident heatmap; no personal data, names, or device identifiers stored.
- FR-9.7** Home location setting: user saves a home GPS coordinate (live fix or manual pin) and a geofence radius (50–500 m, default 100 m); dashboard shows *Monitoring On* when outside the geofence and *Paused – At Home* when inside.

9. System Limitations and Constraints

- LM-1** Android-only target. iOS is excluded due to restrictions on background microphone and sensor access in production.
- LM-2** Background microphone behaviour varies across OEM firmware (MIUI, One UI, ColorOS, EMUI). The Month 1 technical spike will document device-specific behaviour and workarounds.
- LM-3** The Struggle/Resistance class is collected from campus volunteers under simulated conditions, consistent with standard practice in the fall-detection and safety literature. Generalisation to real assault scenarios is acknowledged as a known limitation and reported in the thesis. The audio dataset targets 20–30 volunteers in six real environments; if recruitment falls below 15 by end of Month 2, the augmentation multiplier is raised from $3\times$ to $5\times$ and the generalisability claim is narrowed explicitly.
- LM-4** Camera module is optional. Camera unavailability does not block SOS transmission.
- LM-5** All inference is on-device; no cloud AI is used. Performance depends on the target device's CPU and GPU.
- LM-6** This project is a research prototype. It has not undergone clinical or legal validation as a certified personal safety device.
- LM-7** Indoor and dense-urban GPS accuracy is limited. `FusedLocationProviderClient` mitigates this by combining GPS, Wi-Fi, and cell-tower signals.
- LM-8** Home geofence boundary detection may lag by up to 60 seconds in areas with poor GPS coverage. On a sustained location fix failure the app defaults to active monitoring so safety is never silently lost.
- LM-9** Battery performance of continuous audio monitoring is anticipated to be elevated compared to a standard idle application, as persistent CPU activity, 50 Hz sensor polling, noise cancellation processing, and periodic audio inference all consume device resources. This is a known anticipated risk, not a confirmed outcome. No consumption figure is pre-committed in this proposal. The cascaded pipeline architecture with noise cancellation (FR-3.0) and three-tier VAD gating (FR-3.2–FR-3.4) constitutes the primary architectural mitigation. Empirical benchmarks conducted in Month 1 using Android Battery Historian across three pipeline configurations and three device tiers will establish the viable production architecture and the baseline against which final consumption is reported in the thesis.
- LM-10** The noise cancellation stage (FR-3.0) is optimised for stationary background noise. Performance may degrade in highly dynamic acoustic environments. This limitation is acknowledged and will be assessed in the out-of-distribution evaluation.
- LM-11** SEMOUR is released for non-commercial research use. Its license terms are compatible with this academic FYP and the planned CC BY-4.0 public release of the original Urdu distress recordings; the SEMOUR clips themselves will not be included in the released dataset. License compliance will be documented in the thesis.

10. Data Gathering Approach

Motion Data (Module 2)

Sensor data for the Normal and Fall/Collapse classes will be sourced from the public UCI HAR [1], WISDM [12], MobiAct [23], and UniMiB SHAR [24] datasets. These are direct downloads requiring no original collection effort.

The Struggle/Resistance and Forced Movement classes will be collected from campus volunteers under the following protocol. No public dataset exists for these classes (confirmed systematic search; see Section 5.2), making this collection an original research contribution.

Volunteer recruitment: Target 15–20 volunteers; minimum 50% female; age range 18–45; minimum three different regional accent backgrounds (Punjabi, Pashtun, standard Urdu); no more than three volunteers from the same immediate social group. Recruitment via department notice board and direct team approach across three weeks of Month 2.

Recording equipment and setup: Mid-range Android smartphone in trouser pocket; Physics Toolbox Sensor Suite app (free, Play Store, CSV export); accelerometer and gyroscope at 50 Hz; open campus area or indoor corridor with safety mat for stumble scenarios.

Scenario protocol (per volunteer, 10–15 minutes total):

- Scenario 1 — Wrist grab: second person grabs subject's wrist from behind while walking; subject resists for 5–10 seconds.
- Scenario 2 — Forced directional pull: second person grabs subject's arm and pulls in a different direction against resistance.
- Scenario 3 — Push/stumble: second person pushes subject from the side; subject stumbles without falling.
- Each scenario repeated three times; recording window 10–15 seconds per repetition.

Contingency: If recruitment falls below 15 by end of Month 2, augmentation multiplier raised from 3× to 5× and generalisability claim explicitly narrowed in the thesis.

Audio Data (Module 4)

External corpus (pre-downloaded, Month 1): UrduSER [20] (CC BY 4.0), UrduSpeech [21] (CC BY 4.0, US-Std split), SEMOUR [22] (non-commercial research), CREMA-D [5] (CC BY-SA 4.0), RAVDESS audio-only subset (CC BY-NC 4.0), AudioSet scream subset, ESC-50, and UrbanSound8K are downloaded and integrated before original recording begins, ensuring the project is not blocked by volunteer scheduling. All three Urdu corpora are retained in full before emotion-field filtering to preserve any reclassification flexibility.

Speaker recruitment: Target 20–30 volunteers; minimum 60% female (target user group); age range 18–45; minimum four regional accent backgrounds (Punjabi, Sindhi, Pashtun, standard Urdu); no more than three volunteers from the same immediate social group to avoid vocal

similarity clustering. Recruitment via department notice board, class announcements, and direct approach by all three team members across Weeks 1–2 of Month 2.

Recording environments (six real environments, no artificial substitution):

1. Quiet indoor — university classroom after hours (low noise floor)
2. Indoor ambient — university cafeteria (background chatter)
3. Outdoor quiet — campus lawn (wind, birds)
4. Outdoor busy — main gate / G-9 markaz (traffic, crowds)
5. Indoor reverberant — corridor between classes (echoing)
6. Vehicle interior — car or rickshaw (engine noise, road noise)

Each volunteer records in a minimum of two of the six environments.

Recording protocol (per volunteer, 15–20 minutes total):

- Device: mid-range Android smartphone at 30 cm from mouth, held naturally.
- App: Android AudioRecord API at 16 kHz, 16-bit mono.
- Distress clips: *bachao* $\times 5$, *madad karo* $\times 5$, *choro mujhe* $\times 5$ = 15 clips per volunteer.
- Aggression clips: threatening Urdu speech, 5 clips per volunteer.
- Normal clips: casual Urdu conversation, 5 clips per volunteer.
- Total per volunteer: 25 clips $\times$ ~ 5 seconds each.

Labelling protocol:

- Each clip labelled independently by two annotators from the team.
- Disagreements resolved by the third team member as tie-breaker.
- Inter-annotator agreement measured via Cohen's Kappa; target $\kappa \geq 0.80$.
- Kappa value reported in the thesis as a data quality metric.

Quality control rules:

- Clips below 3 seconds rejected.
- Clips with amplitude clipping (signal reaching maximum) rejected.
- Minimum SNR threshold 10 dB; clips below threshold rejected.
- No class exceeds 40% of the total original Urdu corpus before augmentation.

Ethics and consent:

- Written informed consent form signed by each volunteer before recording.

- Supervisor sign-off as IRB-equivalent per Air University policy.
- Each volunteer assigned an anonymous numeric ID; no personal identifiers stored with audio files.
- Right to withdraw and have all clips deleted, honoured on request at any time.

User Requirements and System Evaluation

User-requirement data will be gathered through structured interviews and a questionnaire with 5–8 women from the target user group to validate confirmation-window design, sensitivity modes, notification format, and onboarding UX.

End-to-end system evaluation will be conducted with 10–15 volunteers on the Air University campus through simulated daily activities and two staged threat scenarios, measuring real-world false-positive rate, detection latency, and confirmation-window usability.

11. Tools and Technologies

Table 2 lists the complete tools and technologies stack for SheAlert, organised by functional area.

Table 2: *Tools and Technologies for SheAlert*

Area	Tool / Technology	Version	Purpose
Programming Language	Python	3.10+	AI model development, training, and augmentation pipeline
ML Framework	PyTorch / TensorFlow	2.x	Multiclass LSTM and YAMNet fine-tuning
Audio Features	Librosa	0.10+	MFCC and Mel-spectrogram generation
Noise Cancellation	SciPy / WebRTC NS	Latest	Spectral subtraction and Wiener filter for FR-3.0 pre-processing stage
Audio Base Model	YAMNet (Google)	TF Hub	Pretrained base for distress classifier
Sensor Model	Multiclass LSTM Classifier	Custom	Four-class motion threat detection
Sensor Datasets	UCI HAR + WISDM + MobiAct + UniMiB SHAR	Public	Normal-activity and fall training data

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Area	Tool / Technology	Version	Purpose
Urdu Audio Corpora	UrduSER + UrduSpeech (Attia) + SEMOUR	Public (CC)	Three complementary Urdu emotional speech corpora spanning broadcast, multi-domain naturalistic, and studio registers
Cross-lingual Audio	CREMA-D + RAVDESS	Public (CC)	Cross-lingual acoustic anchors for fear and anger signatures in Stage 1
Ambient Audio	AudioSet + ESC-50 + UrbanSound8K	Public (CC)	Non-verbal screams and urban ambient audio for Normal and Distress classes
On-device Inference	TensorFlow Lite	2.x	Quantised on-device model deployment
Mobile Framework	Flutter (Dart)	3.x	Complete Android application and UI
Background Audio	Kotlin Foreground Service	API 29+	Persistent background microphone capture
Background CPU	PARTIAL_WAKE_LOCK	API 1+	Keeps CPU running when screen is locked
Tier 1 Audio Filter	WebRTC VAD	Latest	Lightweight voice activity detection
Vision Module	MobileNet SSD	Latest	Lightweight person detection for camera
Primary Alert	Firebase Cloud Messaging	Latest	Real-time SOS push notifications
Fallback Alert	Android SmsManager	API 1+	SMS fallback when no internet
Location	FusedLocation ProviderClient	Latest	GPS + Wi-Fi + cell-tower fusion
Backend Server	FastAPI (Python)	0.100+	Alert forwarding and contact management
Battery Benchmarking	Android Battery Historian	Latest	Empirical battery consumption measurement across pipeline configurations
Battery Optimisation	Battery Optimisations API	API 23+	Whitelist prompt for OEM optimisers

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Area	Tool / Technology	Version	Purpose
Evaluation	scikit-learn	1.x	ROC analysis, F1, precision, recall, Cohen's Kappa
Sensor Recording	Physics Toolbox Sensor Suite	Latest	IMU data collection for volunteer anomaly recordings
IDE	VS Code + Android Studio	Latest	Python, Dart, and Kotlin development

Note on audio corpus architecture: The three Urdu-specific corpora serve distinct and complementary roles. UrduSER provides broadcast-drama naturalistic speech from 10 professional actors validated by native-speaker inter-rater agreement. UrduSpeech adds scale and multi-domain naturalistic coverage with Pakistani-accent-dominant speakers. SEMOUR adds phonetically balanced, studio-quality recordings that anchor the corpus with controlled acoustic conditions. Together these corpora ensure YAMNet is not overfit to any single acoustic register of Urdu before the Stage 2 domain adaptation on assault-specific vocabulary. CREMA-D and RAVDESS serve a different purpose: they provide cross-lingual acoustic grounding for the fear and anger categories, teaching YAMNet the language-independent pitch, energy, and zero-crossing patterns of emotional arousal before any Urdu data is seen.

Note on YAMNet currency: YAMNet (2019) is selected as the base audio model because of its lightweight footprint, mature TensorFlow Lite tooling, and strong public benchmarks on AudioSet [9]. Newer transformer-based audio models (e.g., AST, BEATs) deliver higher accuracy but exceed the on-device latency and battery budget required for continuous monitoring on mid-range Android hardware.

12. Project Stakeholders and Roles

Table 3: Project Stakeholders for SheAlert

Stakeholder	Role and Responsibility
Project Sponsor	Air University, Islamabad, Department of Creative Technologies
Mahad Jokhio (231242)	Student 1, Motion & Sensor AI: UCI HAR, WISDM, MobiAct, UniMiB SHAR dataset preparation; sensor data preprocessing; multiclass LSTM classifier training and evaluation; volunteer anomaly recording coordination; Android sensor data pipeline; threat score streaming; sensor-only fallback mode; TFLite deployment of motion model; placement variability study (pocket vs. bag vs. hand) as original experiment; individual thesis.
Moaz Hassan Khan Manj (231168)	Student 2, Audio + Fusion (Core AI Logic): Urdu distress dataset design, recording, labelling, augmentation; integration of UrduSER, UrduSpeech (Attia), SEMOUR, CREMA-D, RAVDESS, and public ambient datasets; emotion-field filtering of UrduSpeech US-Std split; noise cancellation stage (FR-3.0) implementation; two-stage YAMNet fine-tuning (Stage 1 on full multi-corpus, Stage 2 on original Urdu recordings) and evaluation; MFCC and Mel-spectrogram pipeline; 3-tier VAD cascade; on-device TFLite audio inference; multi-modal fusion engine; threshold tuning and ROC analysis; individual thesis.
Hunain Ahmed (231156)	Student 3, System Intelligence & Validation: Two-stage confirmation window (researched and evaluated as a human-in-the-loop false-alarm reduction system); camera module (evaluated as a third-modality contribution to multi-modal fusion precision); SOS and alert delivery system; FastAPI backend; complete Flutter Android application, heatmap, feedback loop; full system integration; ablation study (formal ownership) measuring the contribution of each modality, the multiclass vs. autoencoder comparison, the noise cancellation contribution, and the confirmation window on false-alarm rate and detection accuracy; individual thesis.
Supervisor	Dr. Samana Batool (Assistant Professor): technical guidance, milestone review, architecture approval, individual thesis evaluation.
Co-Supervisor	Dr. Madiha Yousaf (Assistant Professor): secondary technical guidance and milestone review.
Industrial Supervisor	Maryam Aamir Ashraf (Associate Data Scientist): industry-perspective feedback on model evaluation, deployment readiness, and dataset quality.
FYP Committee	Mid-point evaluation, final defence, grading.
Target Users	Women in Pakistan: structured interviews and usability testing to validate UX design, alert behaviour, confirmation-window interaction, and onboarding flow.

13. Module-Based Work Division

The work distribution below has been finalised and agreed in writing between the three students and the supervisor at the project kick-off meeting. Each student takes formal ownership of a coherent vertical slice of the system, while dataset creation, labelling, testing, and final integration

are explicitly shared responsibilities to ensure equal participation and learning outcomes.

Table 4: Team Member Work Division (Finalised)

Student	Reg. No.	Responsibility / Modules / Features
Mahad Jokhio <i>Student 1, Motion & Sensor AI</i>	231242	Module 1, Motion Sensor Data Pipeline: UCI HAR, WISDM, MobiAct, UniMiB SHAR preparation; sliding-window segmentation and normalisation; Android <code>SensorManager</code> continuous collection in Kotlin. Module 2, Motion Threat Classification: Multiclass LSTM design, training, and evaluation; four-class corpus assembly; volunteer anomaly recording (Struggle/Resistance class); ablation vs. LSTM Autoencoder baseline; TFLite conversion and deployment; sensor-only fallback; threat score stream delivered to Student 3. Original Experiment: Placement variability study (pocket vs. bag vs. hand).
Moaz Hassan Khan Manj <i>Student 2, Audio + Fusion (Core AI Logic)</i>	231168	Module 3, Audio Pipeline and 3-Tier VAD Cascade: Noise cancellation stage (FR-3.0); Foreground Service implementation; WebRTC VAD (Tier 1); rule-based filter (Tier 2); microphone interruption handler; battery benchmarking (Month 1 spike). Module 4, Urdu Dataset and YAMNet Fine-Tuning: Integration of three Urdu corpora (UrduSER, UrduSpeech Attia, SEMOUR) and emotion-field filtering; integration of cross-lingual anchors (CREMA-D, RAVDESS) and ambient datasets; original Urdu recording coordination; labelling protocol and inter-annotator agreement; two-stage YAMNet fine-tuning (Stage 1 on full ~14,000–16,000 clip corpus, Stage 2 on original Urdu recordings); 5-fold cross-validation and OOD evaluation; TFLite conversion; dataset public release. Module 5, Multi-Modal Fusion Engine: Late-fusion formula; context-adaptive weights; threshold tuning and ROC-curve analysis.

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Student	Reg. No.	Responsibility / Modules / Features
Hunain Ahmed <i>Student 3, System Intelligence & Validation</i>	231156	Module 6, Confirmation Window: Researched and evaluated as a human-in-the-loop false-alarm reduction system; Stage 1 and Stage 2 UI; dead man's switch PIN. Module 7, Camera Module: Evaluated as a third-modality contribution to multi-modal fusion precision; MobileNet SSD; fail-safe logic. Module 8, SOS and Alert Delivery: GPS packaging; Firebase FCM; SMS fallback; FastAPI backend; acknowledgement timer. Module 9, Flutter App, UI, Heatmap, Feedback + Backend: Flutter Android application; onboarding; mode configuration; permission monitor; feedback system; heatmap. Full system integration and ablation study (formal ownership) measuring the contribution of each modality, the multiclass vs. autoencoder comparison, the noise cancellation contribution, and the confirmation window on false-alarm rate and detection accuracy.
Shared Responsibility (All Three Students): • Custom dataset recording, labelling, and validation • Testing and evaluation (unit, integration, and end-to-end) • Final integration support • Documentation and report writing <i>"All three team members will collaboratively contribute to dataset creation, labelling, testing, and system validation to ensure equal participation and learning outcomes."</i>		

14. WBS and Gantt Chart

- WBS & Gantt Chart given in **Appendix A**.

15. Mockups

- Mockups are given in **Appendix B**.

16. References

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Appendix A — WBS and Gantt Chart

This appendix contains the Work Breakdown Structure (WBS) and the Gantt Chart for the SheAlert project.

A.1 Work Breakdown Structure (WBS)

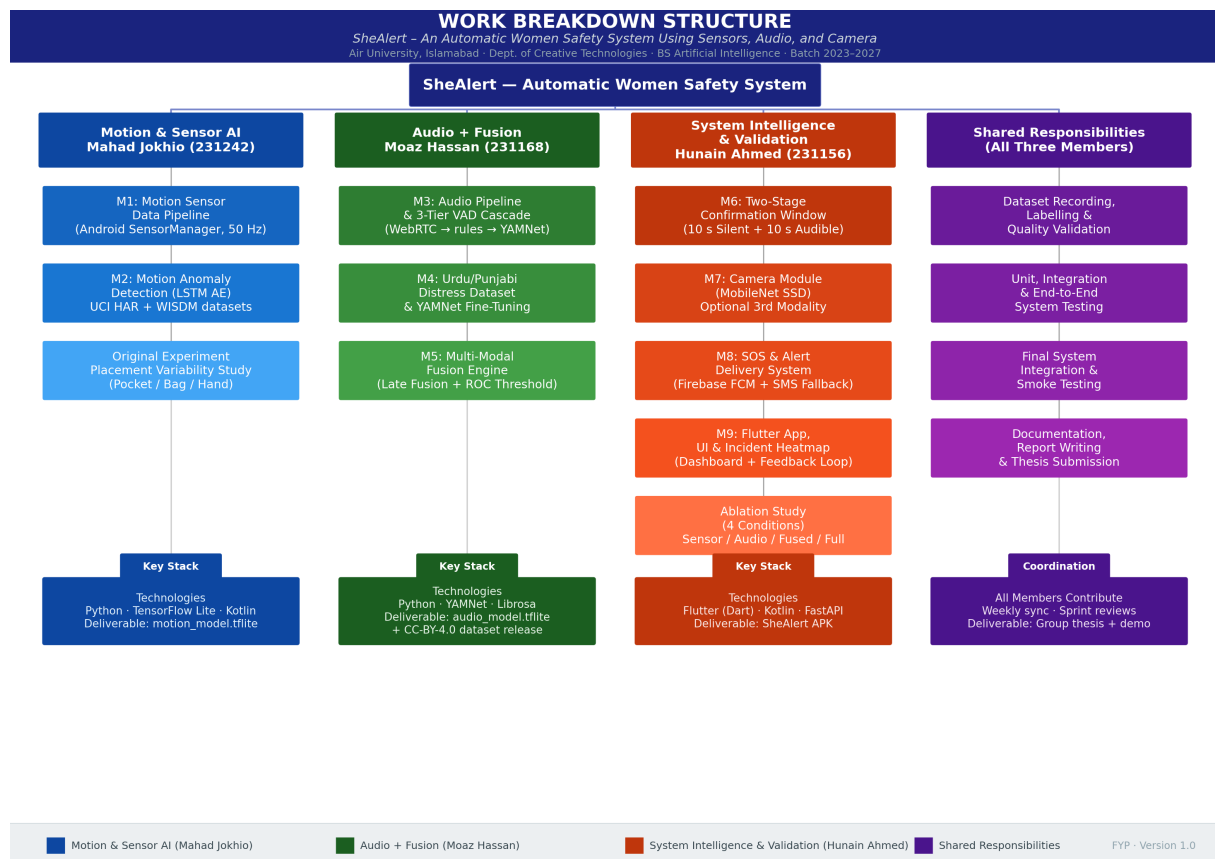


Figure 1: Work Breakdown Structure for SheAlert.

A.2 Gantt Chart

The complete Gantt chart for the SheAlert project spans a 12-month timeline. Due to its size, it has been split into two parts: Part 1 covers the first half of the project (Months 1–6: research, dataset construction, and model development), and Part 2 covers the second half (Months 7–12: integration, evaluation, and final deployment).

A.2.1 Gantt Chart — Part 1 (Months 1–6)

ID	Task	Owner	Dur.	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
P1: Research & Setup															
1.1	Literature review & gap analysis	All	1 mo												
1.2	Architecture finalisation	All	1 mo												
1.3	Tool setup & environment	All	1 mo												
1.4	Mic spike: Android 10/12/14	Moaz	1 mo	★											
P2: Data Preparation															
2.1	UCI HAR & WISDM preprocessing	Mahad	2 mo												
2.2	Urdu/Punjabi audio recording	All	2 mo												
2.3	Augmentation pipeline (1500+ clips)	Moaz	1 mo												
2.4	Volunteer anomaly test recordings	All	1 mo												
P3: Model Development															
3.1	LSTM Autoencoder training & eval	Mahad	3 mo												
3.2	YAMNet fine-tuning & eval	Moaz	3 mo												
3.3	Module evaluation done	All	1 mo												
P4: Real-Time Pipeline															
4.1	TFLite model conversion	All	1 mo												
4.2	Android sensor pipeline (Kotlin)	Mahad	2 mo												
4.3	3-tier audio cascade on Android	Moaz	2 mo												
4.4	Score streams delivered	All	1 mo												
P5: Fusion & Confirmation															
5.1	Fusion engine + adaptive weights	Moaz	1 mo												
5.2	ROC-curve threshold validation	Moaz	1 mo												

Figure 2: Gantt Chart for SheAlert — Part 1 (Months 1–6).

A.2.2 Gantt Chart — Part 2 (Months 7–12)



Figure 3: Gantt Chart for SheAlert — Part 2 (Months 7–12).

Appendix B — Mockups

This appendix contains the wireframe mockups for the six core screens of the SheAlert application. Mockups for Login, Signup, Forgot Password, and About screens are excluded per FYP template guidelines.

B.1 Mockup 1 — Main Dashboard

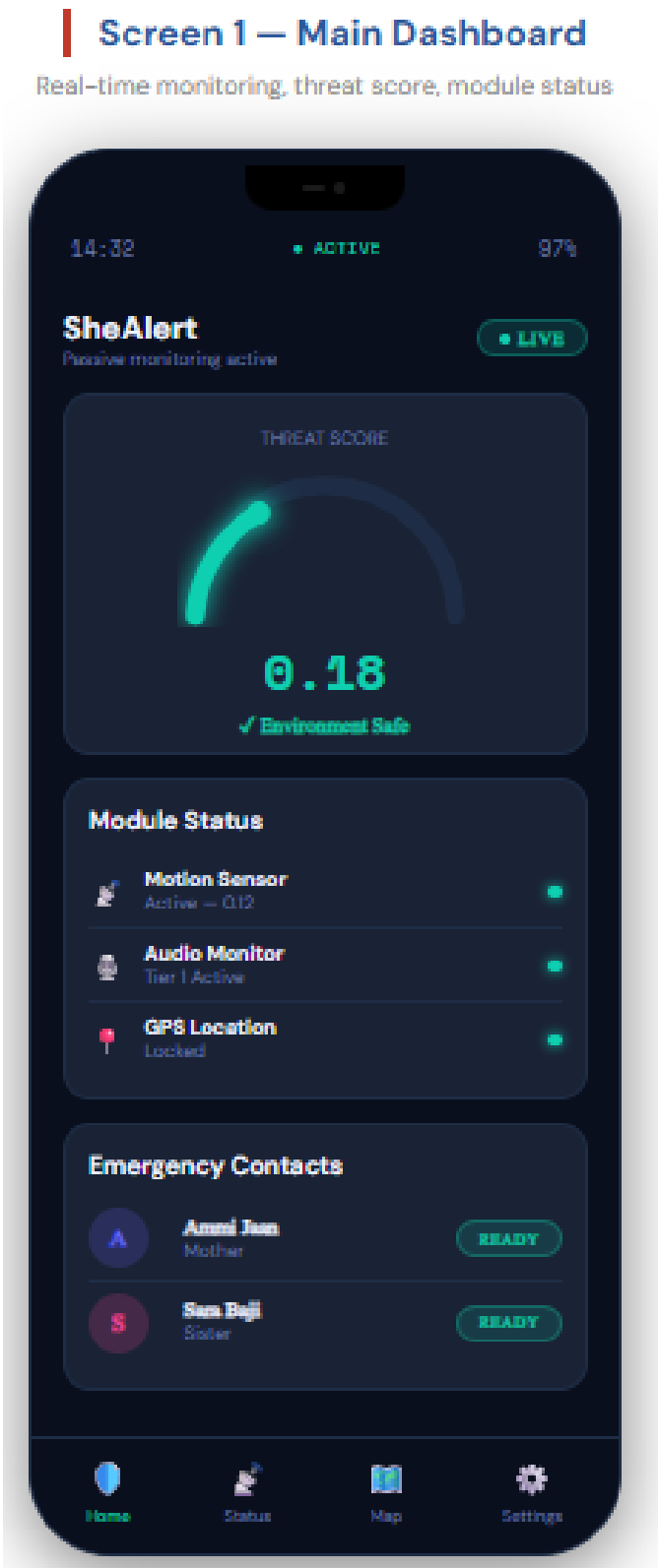


Figure 4: Main Dashboard Screen.

B.2 Mockup 2 — Stage 1 Silent Confirmation Window

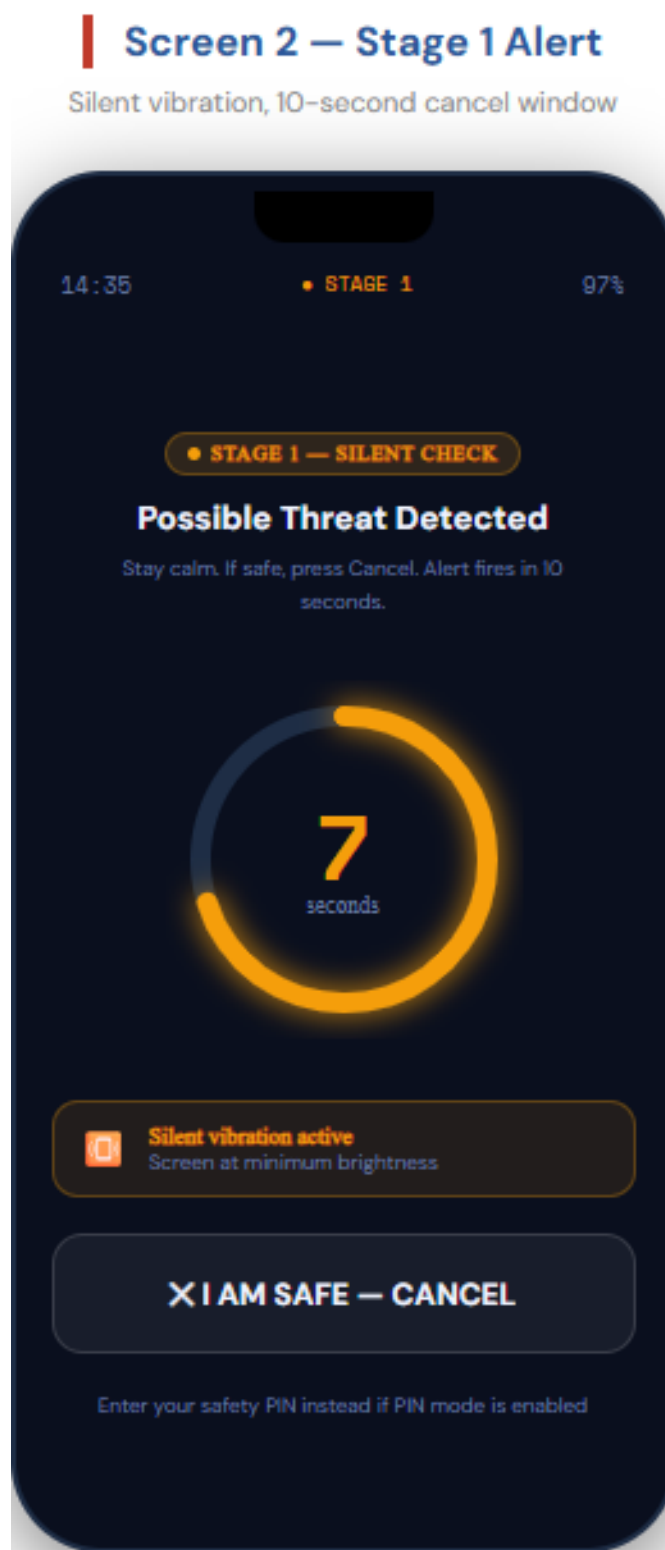


Figure 5: Stage 1 Silent Confirmation Window.

B.3 Mockup 3 — Stage 2 Audible Confirmation Window

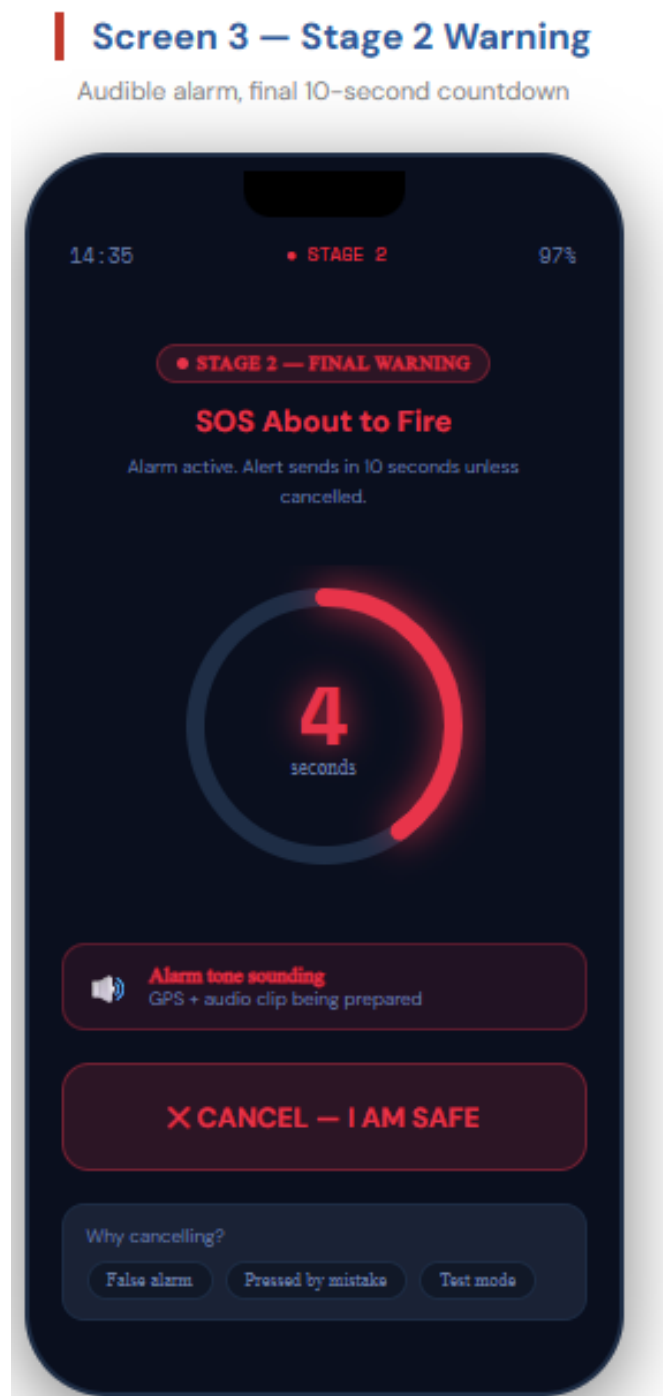


Figure 6: Stage 2 Audible Confirmation Window.

B.4 Mockup 4 — SOS Sent Confirmation

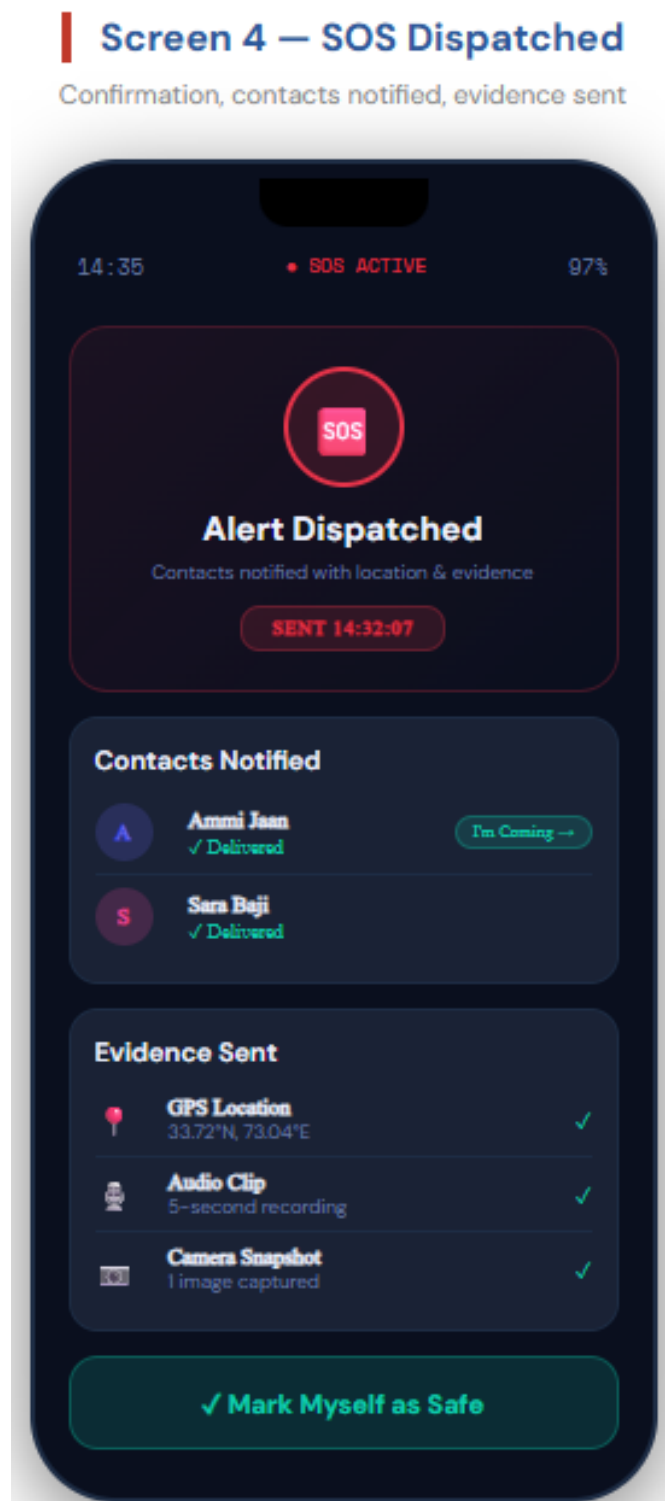


Figure 7: SOS Sent Confirmation Screen.

B.5 Mockup 5 — Incident Heatmap

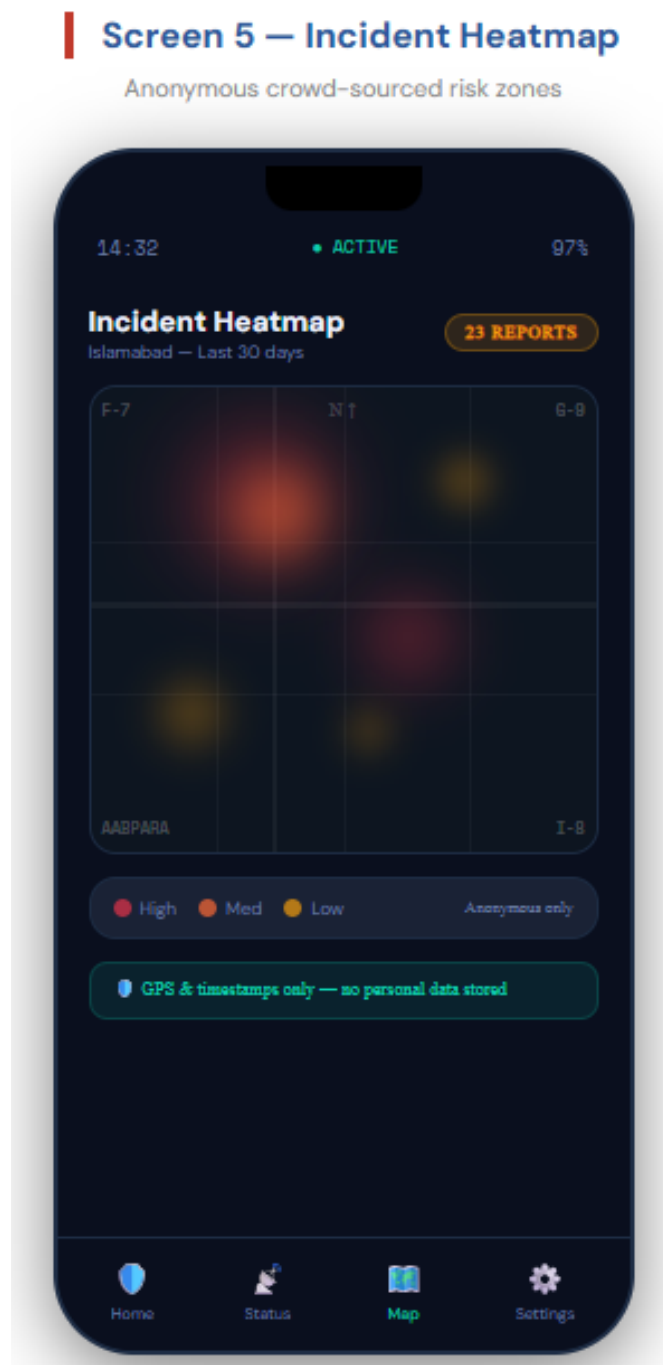


Figure 8: Incident Heatmap Screen.

B.6 Mockup 6 — Settings Screen

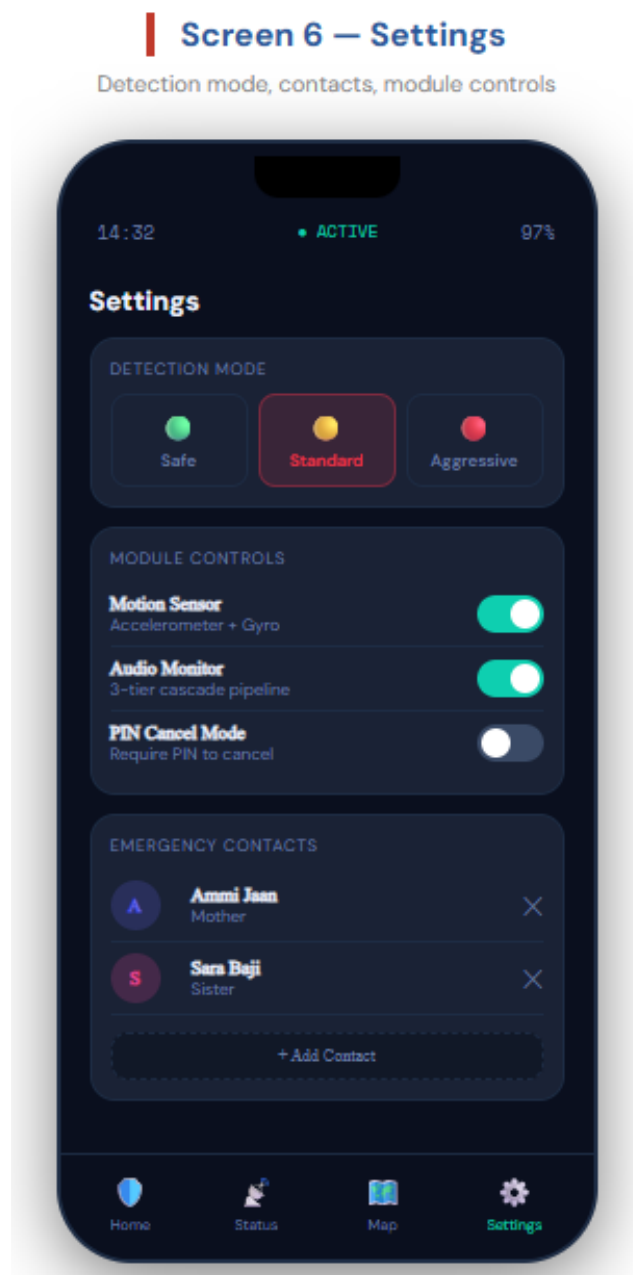


Figure 9: Settings Screen.

Appendix C — Revision Summary (Version 1.0 → Version 2.0)

This appendix documents every change made to the SheAlert scope document in response to the FYP panel's four evaluation comments and the faculty supervisor's additional directives. For each comment the table records the original concern, what was changed, where in the document the change appears, and the detail of the change.

Comment 1 — Training LSTM autoencoder on normal daily activities may produce lots of false positives. It is better to train it as a multiclass problem.

What Changed	Location in Document	Detail
Abstract rewritten	Section 1 (Abstract)	“LSTM Autoencoder for motion anomaly detection” replaced with supervised multiclass LSTM; four named threat classes stated.
Module 2 fully rewritten	Section 8.2, FR-2.1–FR-2.8	Module renamed to “Motion Threat Classification (Multiclass LSTM)”; four classes defined with dataset sources per class; LSTM Autoencoder demoted to ablation baseline (FR-2.7); volunteer recording protocol for Struggle class added (FR-2.8).
OB-1 updated	Section 4.1, OB-1	Target changed from anomaly detection rate to per-class F1-score evaluated after implementation via cross-validation on a subject-disjoint test set across all four classes. An LSTM Autoencoder is retained as a baseline in the ablation study only.
Literature review expanded	Section 5.2	MobiAct and UniMiB SHAR added with confirmed gap for the Struggle class stated and search methodology documented.
Section 4 solution rewritten	Section 4, opening paragraphs	“LSTM Autoencoder” replaced with “supervised multiclass LSTM classifier”; four classes and dataset sources stated; two-stage audio fine-tuning introduced.
Scope updated	Section 7 (Scope)	“Anomaly scoring” replaced with “multiclass threat classification”; noise cancellation stage added to audio layer description.
Stakeholder and work division tables updated	Sections 12 and 13, Tables 3–4	Mahad’s role updated to reflect multiclass classifier; Moaz’s row updated to include noise cancellation stage; Hunain’s ablation row updated to include noise cancellation contribution measurement.
Ablation study updated	FR-5.5	Condition (e) added: multiclass LSTM vs. autoencoder baseline comparison.
Tools table updated	Section 11 (Tools)	“LSTM Autoencoder” replaced with

Comment 2 — The dataset target is very small, and it is difficult to add diversity even with data augmentation.

What Changed	Location in Document	Detail
UrduSER integrated (Version 2.0)	Section 8.4, FR-4.0; Section 16	UrduSER (3,500 clips, CC BY 4.0) [20] added as the primary Urdu emotional speech backbone; class mapping stated (Fear → Distress, Angry → Aggression, Neutral → Normal).
UrduSpeech (Attia) added (Version 2.1)	Section 8.4, FR-4.0; Section 16	UrduSpeech [21] (156 h, CC BY 4.0, arXiv:2605.17846) added as a second Urdu corpus; US-Std split (59.2 h) used, filtered by emotion field to extract fearful/anxious, angry, and neutral clips (~2,000–3,000 relevant clips). Provides large-scale naturalistic multi-domain coverage complementing UrduSER's broadcast register.
SEMOUR added (Version 2.1)	Section 8.4, FR-4.0; Section 16	SEMOUR [22] (15,040 clips, 8 professional Pakistani actors, studio quality) added as a third Urdu corpus; Fearful, Anger, and Neutral classes provide ~5,640 relevant clips. Provides controlled studio recordings that complete the acoustic-register triangle with UrduSER and UrduSpeech.
CREMA-D added (Version 2.1)	Section 8.4, FR-4.0; Section 16	CREMA-D [5] (7,442 clips, 91 actors, CC BY-SA 4.0) added as a cross-lingual acoustic anchor; Fear and Anger subsets (~2,400 clips) teach YAMNet language-independent fear and anger acoustic signatures in Stage 1, substantially expanding RAVDESS coverage.
Public audio datasets added (Version 2.0)	Section 8.4, FR-4.0	AudioSet scream subset, RAVDESS, ESC-50, and UrbanSound8K integrated with class assignments stated.
Original recording target raised	Section 8.4, FR-4.1	Raised from “450+ clips, 8–10 volunteers, 4 environments” to “800–1,000 clips, 20–30 volunteers, 6 real environments.”
Augmentation role reframed	Section 8.4, FR-4.1; OB-2	Augmentation explicitly stated as a regularisation technique only; real diversity is achieved through original recordings, and those now reviewed

Comment 3 — An F1-score ≥ 0.8 on this data split can be achieved, but it does not prove robustness.

What Changed	Location in Document	Detail
Pre-committed score targets removed	OB-1, OB-2, FR-2.3, FR-4.4; Section 1 (Abstract)	All specific “F1-score ≥ 0.80 ” thresholds removed. Replaced throughout with: score targets will be defined after implementation and cross-validation based on dataset characteristics; the team will aim for the highest accuracy achievable; each dataset has its own standard and no threshold is pre-committed.
5-fold cross-validation added	OB-2; FR-2.3; FR-4.4	5-fold speaker-stratified cross-validation mandated; mean F1 $\pm$ standard deviation reporting required in the thesis.
Subject-disjoint split made explicit	FR-2.3; FR-4.4	“Subject-disjoint train/test split mandatory” stated as a hard requirement for both motion and audio models.
OOD environment evaluation added	OB-2; FR-4.4	Out-of-distribution evaluation on one held-out acoustic environment (not seen during training) mandated; generalisation gap to be reported explicitly in the thesis.
Two-stage evaluation reporting	FR-4.4	Stage 1 and Stage 2 fine-tuning results compared in an incremental table in the thesis, providing evidence of domain adaptation benefit.

Comment 4 — Battery consumption will be a serious issue. Fixing a solution even before starting the project is not a good approach.

What Changed	Location in Document	Detail
Battery reframed as anticipated risk	Section 3.3; Section 1 (Abstract)	Battery overconsumption reframed as an anticipated — not confirmed — challenge. The specific reasons are stated: persistent CPU activity from the background service, 50 Hz sensor polling, noise cancellation processing, and periodic audio inference. The cascaded pipeline with noise cancellation is named as the mitigation strategy. No consumption figure is pre-committed.
Pre-committed consumption figure removed from FR-3.4	Section 8.3, FR-3.4	“Target battery drain 3–5% per hour” removed. Replaced with: battery consumption is anticipated to be lower than always-on inference due to cascaded gating; empirical benchmarking in Month 1 will confirm or revise this expectation.
OB-5 updated	Section 4.1, OB-5	Added explicit statement that elevated battery consumption is anticipated due to background processing; the cascaded pipeline and noise cancellation are the primary architectural mitigations; production architecture selected from empirical results only.
LM-9 rewritten	Section 9, LM-9	Now explicitly states battery performance is anticipated to be elevated with named reasons; cascaded architecture with noise cancellation named as mitigation; empirical Month 1 benchmark gates final architecture decision.
Noise cancellation stage added (FR-3.0)	Section 8.3, FR-3.0 (new)	A dedicated noise cancellation pre-processing stage added before all VAD tiers, using spectral subtraction and a Wiener filter. This stage both improves classification accuracy and reduces spurious Tier-2 activations, contributing to the battery mitigation strategy. FR-3.2 and FR-3.3 updated to operate on the noise-cancelled signal.

All other sections — Vision Statement (Section 6), Module 1 (Section 8.1), Modules 6–9 (Sections 8.6–8.9), WBS/Gantt (Section 14), Mockups (Section 15), Appendix A, and Appendix B — are unchanged from Version 1.0. Version number updated to 2.1 on cover page and footers throughout.